

How the Teaching Style and Interpretation Type of State Interventions Shape Multi-Agent Coordination

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Abstract

How can an external teacher accelerate coordination among multiple learning agents? We investigate this question in a multi-agent foraging task where a simulated teacher can physically relocate agents—a direct but ambiguous pedagogical signal. We formalize a taxonomy of teaching styles (e.g., *Undoing*, *Correcting*) and agent interpretation rules. Our computational experiments reveal that the *Undoing* style is a **uniquely powerful catalyst for spatial coordination**, most effectively breaking behavioral symmetry between two agents by imposing spatial constraints that define “home regions.” This catalytic effect scales, reinforcing niche specialization in three-agent systems. However, success is governed by a **pedagogical matching principle**: interventions fail when the agent’s interpretation rule mismatches the teacher’s intent (e.g., *Correcting* fails when interventions are interpreted as punitive). Our work provides a formal framework showing how structured guidance and agent interpretation jointly shape the emergence of collective organization.

Keywords: Multi-agent reinforcement learning; pedagogy; state intervention; coordination; spatial specialization.

Introduction

Teaching in complex, dynamic environments often extends beyond verbal instruction to include direct manipulation of the learner’s state. In multi-agent settings, the challenge of such pedagogical intervention intensifies: the teacher must guide individual learning while fostering collective coordination under real-time resource competition. While previous research has explored how agents interpret physical corrections in single-agent tasks, the role of state interventions as a tool for shaping group-level behavior remains under-explored.

This paper investigates a core computational question: **How can different algorithmic teaching styles—defined by when and where an agent is moved—catalyze and stabilize spatial coordination in multi-agent systems?** Unlike traditional reinforcement learning where coordination emerges slowly from self-interested exploration, we treat state intervention as a form of fast, asymmetric environmental shaping. We specifically examine how interventions can break the initial behavioral symmetry between agents, leading to *spatial specialization*—a division of labor that minimizes competitive overlap.

The complexity of this teaching scenario is defined by delayed pedagogical outcomes and the teacher’s need to manage multiple agents under spatial uncertainty. Rather than focusing on the agent’s internal interpretive ambiguity, we adopt a **system-level modeling approach** to map the performance

landscape across various teaching styles and agent interpretation rules.

To our knowledge, this is the first systematic investigation of state interventions in real-time, multi-agent, discrete environments where the teacher lacks access to the learner’s feature representation. By bridging the gap between heuristic pedagogical strategies and multi-agent coordination, our work provides a computational foundation for designing adaptive agents that learn effectively from structured external guidance. Our contributions are as follows:

- We formalize a taxonomy of algorithmic teaching styles designed to shape multi-agent coordination through state interventions.
- We identify a **pedagogical matching principle**, demonstrating that the effectiveness of an intervention depends on the alignment between the teaching strategy and the agents’ interpretation rules.
- We demonstrate that specific interventions act as uniquely powerful catalysts for symmetry breaking and stable spatial niche formation.

Prior work

Our work sits at the intersection of computational models of teaching, reinforcement learning from human feedback, and multi-agent systems. This section provides an overview of the relevant literature along these axes, highlighting how our approach diverges from and extends prior research.

Computational Models of Teaching and Intervention

Formal models of teaching often build upon the framework of Partially Observable Markov Decision Processes (POMDPs) to capture the interactive dynamics between a teacher and a learner (Kaelbling, Littman, & Cassandra, 1998). This formalism has been productively applied to understand human pedagogical reasoning (Shafto, Goodman, & Griffiths, 2014) and to design algorithms for efficient instruction (Rafferty, Brunskill, Griffiths, & Shafto, 2016). A key insight from this line of work is that teaching is not merely reinforcement; it is a communicative act where signals are chosen to influence the learner’s beliefs and future behavior (Ho, Cushman, Littman, & Austerweil, 2019, 2021).

Physical state interventions encode a direct yet ambiguous teaching signal. Prior work explored how people use interventions to guide an agent to a goal while avoiding obstacles (Ho, Littman, & Austerweil, 2017). More recently, Losey, Bajcsy, O’Malley, and Dragan (2022) formalized physical human-robot interaction as a POMDP where the robot maintains a belief over the human’s unknown reward function, that is updated through physical corrections interpreted as intentional signals. Their work demonstrates that heuristic interpretations of interventions (e.g., updating one feature weight at a time) can improve learning efficiency. However, these approaches are designed for **single-agent, continuous-control tasks** (e.g., robotic manipulation) and assume a **shared, known feature space** between human and robot, which allows for a relatively straightforward mapping from physical corrections to reward function updates.

In contrast, our work investigates discrete grid-based environments where agents learn *tabula rasa* via model-free RL without prior knowledge of task features or reward structure. The teacher’s interventions (e.g., dragging an agent to a new tile) must therefore be interpreted through heuristic rules defined at the level of state transitions and value updates, not feature-weight adjustments. We extend the idea of heuristic interpretation by proposing and systematically evaluating a broader taxonomy of such rules.

Reinforcement Learning from Human Feedback and Guidance

The paradigm of Reinforcement Learning from Human Feedback (RLHF) and related interactive techniques provide a rich foundation for learning from various forms of human input (Luo, Xu, Wu, & Levine, 2024). In such frameworks, human interventions are often treated as demonstrations or preference labels. For instance, Luo et al. (2024) treat human interventions during high-dimensional manipulation tasks as *suggestions*, where the agent learns directly from the state-action pair induced by the human.

A crucial distinction in our setting is the nature of the task and agent embodiment. Unlike continuous control tasks where a human can kinesthetically guide an end-effector through a physically plausible trajectory, our discrete environment introduces a fundamental asymmetry: the teacher can *teleport* an agent to any tile. An action the agent cannot replicate on its own. This breaks the direct “*suggestion as demonstratable action*” assumption common in RLHF for robotics. Consequently, we must consider interpretations beyond direct imitation, such as treating an intervention as a reset, an interruption, or even a punitive signal.

Multi-Agent Learning and External Shaping

In multi-agent reinforcement learning (MARL), coordination often emerges from self-interested learning (Axelrod, 1984; Austerweil et al., 2015) or is engineered through specific reward structures (Wolpert & Tumer, 2001) and communication protocols (Foerster, Assael, De Freitas, & Whiteson, 2016). External shaping by a teacher adds a new dimension

to this problem. While there is extensive literature on centralized training for decentralized execution (Lowe et al., 2017; Rashid et al., 2020; Foerster, Farquhar, Afouras, Nardelli, & Whiteson, 2018) and on learning communication protocols (Foerster et al., 2016), the explicit study of how **an external teacher can use state interventions to shape multi-agent coordination** is nascent.

Classic game-theoretic analyses show that cooperation can emerge in repeated interactions without explicit social motivations (Axelrod, 1984), and self-interested agents can discover cooperative strategies in many grid-world games (Austerweil et al., 2015). However, analyzing two or more agents with continuous-time, interactive signals remains open. Our work investigates how targeted state interventions (a form of fast, asymmetric environmental shaping) can accelerate and stabilize the emergence of coordinated behaviors, such as division of labor, in a learning population. This connects to broader questions in social learning about how environmental structure mediates the value of external information (Wu et al., 2025). We contribute by providing a formal framework to study this specific shaping mechanism.

Method

We investigate a real-time foraging task in a discrete grid world. A fixed set of tiles are designated as spawn tiles, where pellets appear stochastically over time following a Poisson process (Kingman, 1992). We focus on a *smooth* distribution, where spawn tiles are grouped into a small number of contiguous, spatially coherent regions (Wu et al., 2025). Agents occupy a single tile and move continuously toward adjacent tiles (including diagonals) over a fixed duration. When an agent overlaps with a pellet, it receives a reward R_{pellet} and the pellet is consumed.

A *simulated teacher* with perfect knowledge of the spawn distribution (but not real-time pellet presence) can perform *state interventions*: forcibly relocating an agent to a new tile at any moment. The agent then resumes its policy from this new state. The teacher decides when to intervene by estimating, for each possible action an agent could take from its current position, the long-term value of that action based solely on the known spawn distribution. If the agent’s chosen action is estimated to be suboptimal (i.e., less valuable than the best available alternative from that position), the teacher triggers an intervention with a probability specified by the *intervention rate* parameter. The intervention itself is determined by the *teaching style*.

This setup introduces several learning challenges: (1) rewards are dynamic and stochastic; (2) agents learn online via reinforcement learning with no prior knowledge; (3) we analyze systems involving **two or three agents** operating concurrently, which creates competitive pressure and requires spatial coordination; (4) interventions provide an ambiguous, real-time pedagogical signal that agents must interpret within these multi-agent social dilemmas. This forms a complex testbed for evaluating how state interventions affect collec-

tive coordination and competition in dynamic environments.

Intervention styles

Our environment is modeled as a Markov Decision Process (MDP) (S, A, T, R, γ) , where the state space S corresponds to the agent’s discrete grid position; the action space A includes movements to the eight adjacent tiles; the transition function T is deterministic; the immediate reward $R(s_t, a_t, s_{t+1})$ is stochastic, depending on whether a pellet is collected during the movement from s_t to s_{t+1} ; and γ is the discount factor.

Agents learn via Q-learning (Sutton & Barto, 2018) with an ϵ -greedy policy. The Q-update for (s_t, a_t, s_{t+1}, r_t) is:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \quad (1)$$

A core challenge is how an agent should update its value function when its movement is interrupted by a state intervention. We formalize this by distinguishing between:

- s_{t+1}^A : the state the agent would have reached autonomously,
- s_{t+1}^I : the agent’s state post-intervention,
- a_t : the original action attempted by the agent,
- r_t : environmental reward received prior to interruption.

We define a^I as the action from s_t that best reduces distance to s_{t+1}^I , and r^I as the reward obtained at s_{t+1}^I . To systematically

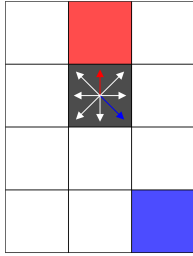


Figure 1: As an agent moves along the red arrow (action a_t) moving from the black (s_t) to red tile (s_{t+1}^A), it is intervened on and is placed on the blue tile (s_{t+1}^I). The blue arrow a^I is the closest action from the starting black tile s_t the post-intervention blue tile s_{t+1}^I .

evaluate these interpretation types, we define four algorithmic **teaching styles** that determine *when* and *where* to intervene:

- **Undoing**: Returns the agent to its previous state s_t .
- **Correcting**: Moves the agent to a randomly selected pellet spawn tile within a 5×5 neighborhood of s_t .
- **Exploration-encouraging Correcting**: Moves the agent to a non-spawn tile adjacent (within a 3×3 area) to a spawn tile in the 5×5 neighborhood of s_t .
- **Restart**: Moves the agent to a non-spawn tile in the 5×5 neighborhood of s_t , far from any spawn tile.

An intervention is triggered probabilistically (via an *intervention rate* parameter) when the agent’s current action is suboptimal relative to the known spawn tile distribution.

Intervention Interpretation

How should an agent update its value function when its movement is interrupted by a state intervention? When externally displaced, an agent must infer the pedagogical intent behind the intervention. We propose six distinct **interpretation types** for mapping an intervention event to a Q-update:

- **Suggestion**: Treats s_{t+1}^I as a recommended goal. *Update*: $(s_t, a^I, s_{t+1}^I, r^I)$.
- **Reset**: Treats intervention as a neutral relocation adapted from (Ho et al., 2017). *Update*: $(s_t, a_t, s_{t+1}^A, r_t)$.
- **Interrupt**: Treats intervention as a mere interruption adapted from (Ho et al., 2017). No Q-update is performed.
- **Transition**: Treats intervention as a positive guided transition. *Update*: $(s_t, a^I, s_{t+1}^I, +2R_{\text{pellet}})$. If $s_{t+1}^I = s_t$, it is treated as negative: $(s_t, a_t, s_{t+1}^A, -2R_{\text{pellet}})$.
- **Disrupt**: Treats any intervention as inherently negative. *Update*: $(s_t, a^I, s_{t+1}^I, -2R_{\text{pellet}})$.
- **Impede**: Treats intervention as a punishment for the original action. *Update*: $(s_t, a_t, s_{t+1}^A, -2R_{\text{pellet}})$.

These interpretation types represent distinct computational hypotheses for how physical interventions function as communicative acts within the learning process.

Policy Evaluation and Coordination Metrics

The stochastic and dynamic nature of pellet generation, and multi-agent interaction violates the standard Markov assumption. Explicitly modeling pellet counts would lead to a combinatorial state explosion. Therefore, to evaluate and compare policies, we approximate the **expected reward** of a policy π through simulation:

$$\hat{V}(\pi) = \frac{1}{100} \sum_{k=1}^{100} \sum_{\tau=0}^{30} \gamma^\tau r_k^{(\tau)}, \quad (2)$$

where $r_k^{(\tau)}$ is the reward obtained at step τ in the k -th rollout. Each rollout starts the agent from a random position and executes π for 30 steps in a new environment instance (with other agents). This metric estimates the agent’s average foraging efficiency under environmental and interactive stochasticity.

To systematically map the multi-agent coordination landscape, we perform a factorial analysis across the proposed taxonomy of *teaching styles* and *interpretation types*. This approach allows us to investigate how specific combinations of pedagogical signals and agent-level heuristics interact with environmental topology. By evaluating these configurations in spatially continuous “smooth” environments, we aim to

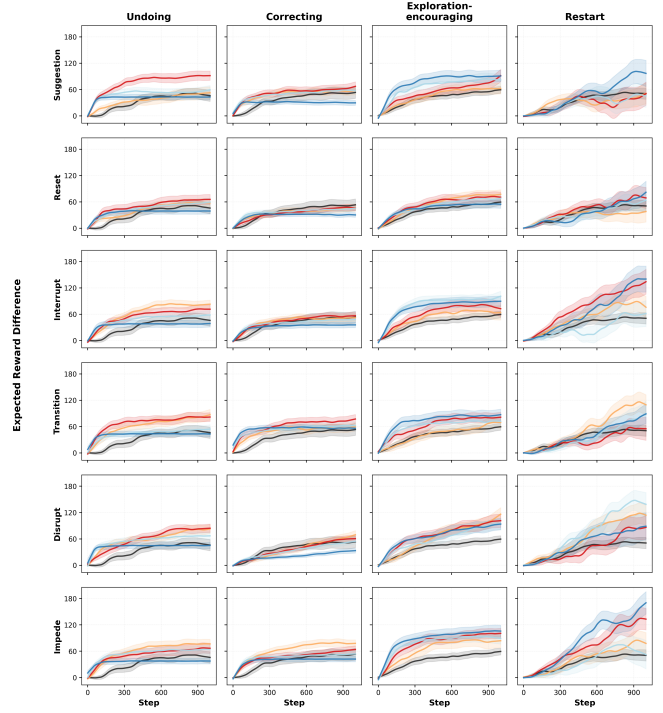
identify the computational conditions that yield the most stable policy acquisition and group-level performance.

Coordination emerges as a critical challenge, when multi-agent systems scale. Ideally, agents should minimize competitive overlap by partitioning the environment—a phenomenon we term *spatial specialization*. The goal is to investigate how intervention styles influence coordination.

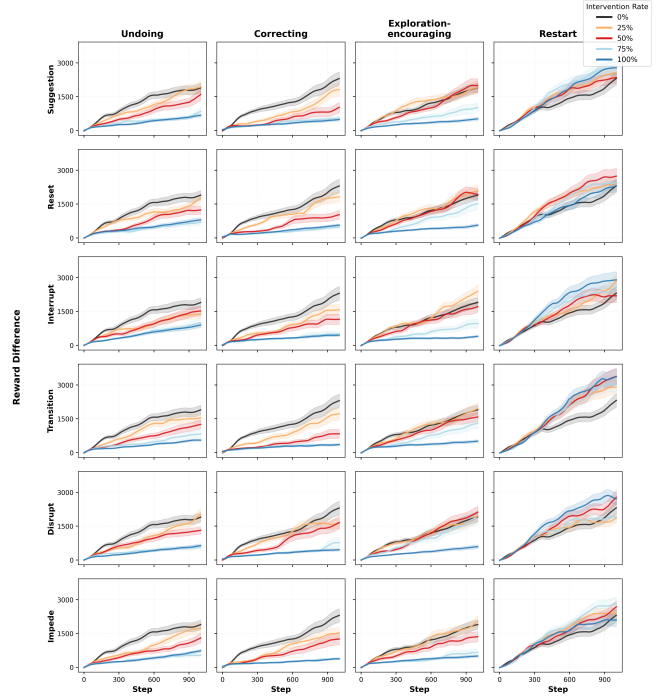
Results and Analysis

Emergence of Coordination in Two-Agent Systems In an environment with two pellet-generating regions, we measure *fair coordination* through two metrics: (1) **Policy Divergence**: The degree to which agents specialize in different regions, measured by the difference in their “first-region preference” (Expected Reward in Region 1 minus Region 2). Larger values indicate greater spatial differentiation. (2) **System Equilibrium**: The balance of resource acquisition, measured by the absolute score difference between agents. Smaller values indicate fairer resource distribution. A factorial ANOVA on Policy Divergence revealed significant main effects of teaching style ($F(3,4704) = 418.96, p < .001$), interpretation type ($F(5,4704) = 9.03, p < .001$), and intervention rate ($F(3,4704) = 8.88, p < .001$), as well as significant teaching style $\times$ interpretation type ($F(15,4704) = 2.86, p < .001$) and teaching style $\times$ rate ($F(9,4704) = 6.26, p < .001$) interactions. For System Equilibrium, teaching style ($F(3,4704) = 267.02, p < .001$) and intervention rate ($F(3,4704) = 82.09, p < .001$) were significant, while interpretation type alone was not ($F(5,4704) = 1.56, p = .17$); however, its interaction with teaching style was significant ($F(15,4704) = 2.99, p < .001$), indicating that interpretation effects depend on the teaching context. Figures 2a and 2b present these results. While all teaching styles improved Policy Divergence relative to no intervention, their effects on System Equilibrium diverged sharply. The *Restart* style yielded significantly worse equilibrium than all other styles ($t = 21.98, p < .001, d = 0.87$; $M_{\text{Restart}} = 2626$ vs. $M_{\text{others}} = 1167$), indicating that abrupt relocations to non-productive areas destabilize fair resource sharing. The *Undoing* style emerged as the most *balanced* intervention: it produced moderate Policy Divergence ($M = 74.0$) while maintaining strong System Equilibrium ($M = 949.8$). Although *Restart* generated the highest raw Policy Divergence ($M = 165.7$), this came at the cost of severe equilibrium degradation. Thus, *Undoing*—by returning agents to their previous positions—effectively penalizes encroachment on suboptimal paths without introducing the destabilizing effects of more aggressive relocations. The *Correcting* style, which redirects agents toward nearby spawn tiles, required high intervention rates (≥ 0.75) to achieve both objectives. At high rates, *Correcting* produced significantly better System Equilibrium than at low rates ($t = 12.26, p < .001, d = 0.71$; $M_{\text{high}} = 478$ vs. $M_{\text{low}} = 1421$), suggesting that consistent corrective feedback is needed to ensure balanced outcomes.

Turning to *interpretation types*, the *Reset* rule—which



(a) Policy Divergence. Larger values reflect better coordination.



(b) System Equilibrium. Smaller values reflect greater fairness.

Figure 2: Policy Divergence and System Equilibrium for Two-Agent Experiments. Rows and columns vary in intervention interpretations and teaching styles, respectively. Error bands encode standard error. An appropriate teaching style accelerates coordination among agents, allowing each agent to focus on a specific region with balanced scoring.

treats interventions as neutral relocations without updating action values—performed significantly worse than other interpretations across teaching styles ($t = -5.53$, $p < .001$, $d = -0.19$; $M_{\text{Reset}} = 86.3$ vs. $M_{\text{Others}} = 103.3$). More strikingly, the *Disrupt* interpretation showed a pronounced mismatch with the *Correcting* style: when agents interpreted corrective relocations as punitive signals, Policy Divergence dropped sharply ($t = -13.12$, $p < .001$, $d = -0.77$; $M_{\text{Disrupt+Correcting}} = 58.9$ vs. $M_{\text{Disrupt+Others}} = 125.9$). This pedagogical mismatch—where well-intentioned guidance is misread as punishment—actively undermines the differentiation of labor. These findings highlight a key **pedagogical matching principle**: successful coordination requires not only appropriate teaching actions but also compatible learner interpretations. When the agent’s heuristic aligns with the teacher’s intent (e.g., treating corrections as suggestions rather than disruptions), interventions accelerate spatial specialization; when they conflict, even beneficial guidance can impede coordination.

Niche Reinforcement in Three-Agent Systems In an environment with three regions and three agents, we measured *spatial attachment*—the degree to which agents concentrate their visits within their nearest region—as an index of emergent niche specialization. A factorial ANOVA revealed that all experimental factors significantly influenced Region Attachment: teaching style ($F(3, 4704) = 8517.27$, $p < .001$), interpretation type ($F(5, 4704) = 55.92$, $p < .001$), and intervention rate ($F(3, 4704) = 455.05$, $p < .001$). All two-way and three-way interactions were also significant ($p < .001$), indicating that the effectiveness of any pedagogical strategy depends jointly on how interventions are delivered and how agents interpret them. Figure 3 illustrates these patterns. The *Undoing* and *Correcting* styles produced dramatically higher spatial attachment than the other styles ($t = 100.37$, $p < .001$, $d = 2.90$; $M_{\text{Undoing+Correcting}} = 586.6$ vs. $M_{\text{Others}} = 208.2$). Examining styles individually, *Correcting* yielded the highest attachment ($M = 607.9$), followed closely by *Undoing* ($M = 565.2$), while *Exploration-encouraging Correcting* ($M = 299.3$) and *Restart* ($M = 117.1$) lagged considerably behind. These results confirm that interventions which guide agents toward productive locations—or constrain them to previously visited states—function as a *positive feedback loop* for spatial coordination, implicitly solidifying each agent’s role within an emerging division of labor. Intervention rate amplified these effects, but not uniformly across styles. Correlation analyses revealed significant positive relationships between rate and Region Attachment for *Undoing* ($r = .65$, $p < .001$), *Correcting* ($r = .48$, $p < .001$), and *Exploration-encouraging Correcting* ($r = .38$, $p < .001$). Critically, *Restart* exhibited a *negative* correlation ($r = -.24$, $p < .001$): higher intervention rates further *reduced* spatial attachment, actively dismantling emerging niches rather than reinforcing them. This divergent pattern underscores that more intervention is not universally beneficial—its effect depends entirely on the pedagogical content of the relo-

cation. The *Exploration-encouraging Correcting* style produced a more diffused spatial footprint ($M = 299.3$); while agents remained primarily in their assigned regions, they exhibited higher cross-regional foraging than under *Undoing* or *Correcting*. This confirms that this style prioritizes individual policy robustness—exposing agents to a broader state space—over strict group coordination. The *Restart* style proved unambiguously detrimental to spatial coordination ($t = -108.51$, $p < .001$, $d = -2.29$; $M_{\text{Restart}} = 117.1$ vs. $M_{\text{Others}} = 490.8$). By relocating agents to non-productive tiles far from spawn regions, *Restart* erased the spatial structure that other styles cultivated, leading to chaotic, uncoordinated foraging. The massive effect size ($d = 2.29$) highlights that this is not a marginal difference but a qualitative failure of the pedagogical strategy.

Discussion

This study systematically explores how algorithmic teaching styles, mediated by agent interpretation rules, can catalyze spatial coordination in multi-agent reinforcement learning. Our central finding is that the ***Undoing* intervention style serves as a uniquely powerful catalyst for multi-agent coordination**, most effectively breaking behavioral symmetry between two agents. The overall effectiveness of pedagogical intervention, however, is governed by a **pedagogical matching principle**: successful coordination requires coherence between the teacher’s external action and the learner’s internal heuristic. Our results provide concrete evidence for this principle by revealing critical *mismatches*. The *Reset* interpretation consistently led to poor performance across all teaching styles, indicating a universal failure to align with pedagogical intent. More specifically, a severe mismatch occurred when agents using the *Disrupt* interpretation (viewing all interventions as negative) were subjected to the *Correcting* style, which actively undermined the very coordination it aimed to promote.

Crucially, the coordination-promoting effect of appropriate teaching scales to more complex multi-agent societies. In three-agent systems, both *Undoing* and *Correcting* styles robustly reinforced the emerging spatial niches, significantly increasing agents’ attachment to their primary regions. This demonstrates that targeted state interventions can transcend simple pairwise coordination and act as a *positive feedback loop* that solidifies collective organization. The failure of the *Restart* style across both two- and three-agent scenarios further underscores that not all interventions are helpful; pedagogical success requires the external guidance to be structured in a way that is computationally interpretable and aligned with the goal of reducing competitive overlap.

These requirements for interpretability and goal alignment are met by the distinct pedagogical policies that our teaching styles operationalize. The *Undoing* style, which forcibly returns the agent to its previous state, emerged as a uniquely powerful catalyst for multi-agent coordination. Its efficacy does not stem from conveying a specific reward or penalty,

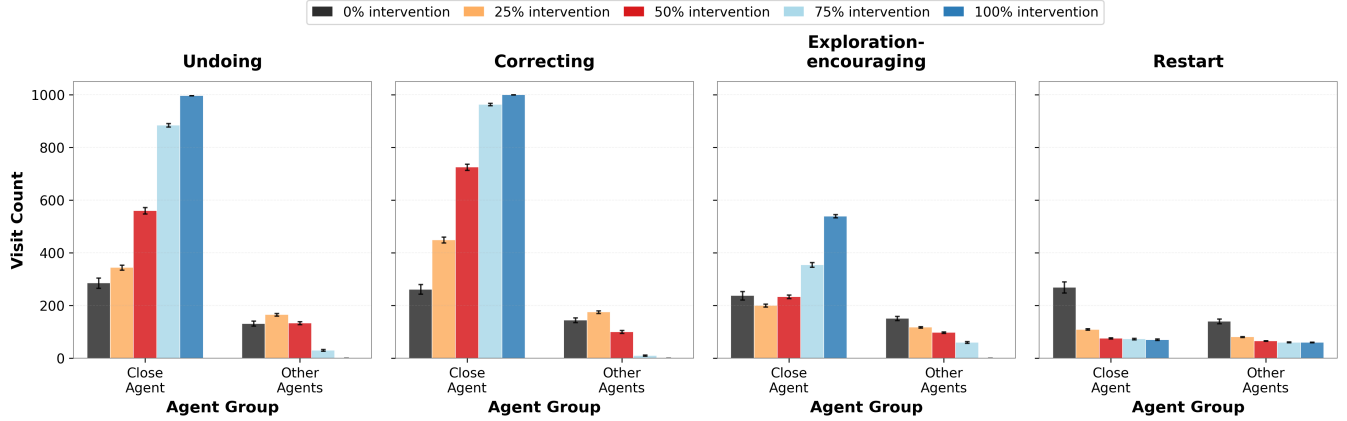


Figure 3: Three-agent simulation results. Visit count of agents for different teaching styles. An appropriate teaching style can enhance the coordination among multiple agents.

but from imposing a strong *spatial constraint*. By repeatedly resetting an agent’s position, it effectively defines a “home region” and increases the cost of venturing into areas dominated by others. This mechanism rapidly breaks behavioral symmetry and enforces spatial niche formation—a crucial step in solving the “clash of interests” in collective learning. This finding demonstrates how structured external intervention can *accelerate and stabilize* the emergence of cooperative equilibria that self-interested learners might otherwise discover slowly or unreliably (Austerweil et al., 2015).

Conversely, the *Correcting* style functions more as a **stabilizing force**. Once a basic spatial division has emerged, *Correcting* is sufficient to reinforce these existing niches by guiding agents back to their respective optimal regions without the high cost of total displacement.

However, its effectiveness as a stabilizer is contingent upon the agent’s interpretation, as evidenced by its severe incompatibility with the *Disrupt* rule. This observation refines the pedagogical matching principle: a well-intentioned guiding correction can fail if the learner misinterprets it as a cautionary demonstration.

Together, these insights suggest a potential trajectory for adaptive pedagogical systems: transitioning from symmetry-breaking interventions (*Undoing*) to refinement-oriented guidance (*Correcting*) as the multi-agent system matures, while remaining mindful of the agents’ evolving interpretative frameworks.

Future Work

The current study establishes that state interventions, particularly the *Undoing* style, can act as a powerful catalyst for spatial coordination under specific conditions (smooth resource distributions, tabula rasa learners). A critical trajectory for future research lies in testing the boundary conditions and generalizability of these effects by introducing greater ecological complexity.

First, transitioning from fully observable grid worlds to Partially Observable Markov Decision Processes (POMDPs)

would introduce realistic perceptual uncertainty. In such settings, state interventions could function not only as spatial corrections but also as *information-shaping signals* that reduce a learner’s belief entropy about the reward landscape. The teacher’s role would thus evolve from trajectory correction to active epistemic guidance, where interventions like *Undoing* might serve to prune sub-optimal hypotheses or direct attention. Exploring whether the catalytic effect of *Undoing* persists, or how the required **pedagogical matching** between style and interpretation changes under partial observability, is a key next step.

Second, the robustness of our identified mechanisms across different environmental structures remains to be explored. For instance, would *Undoing* remain the dominant catalyst in environments with fragmented or dynamically shifting resources? Investigating how the effectiveness of different teaching styles varies with environmental topology will clarify the scope of our findings and inform the design of context-aware pedagogical agents.

Furthermore, the scalability and inherent heterogeneity of multi-agent systems present a vital frontier. Beyond the small, homogeneous groups explored here, larger populations introduce complex collective dynamics with diverse learning rates, sensory capabilities, or intrinsic motivations. Future work should focus on how a teacher (centralized or distributed) can use state interventions to manage such a “heterogeneous classroom”—enforcing specialization, preventing maladaptive cascades, and maintaining a robust socio-technical equilibrium. This entails understanding how to balance individual learning efficiency with global stability as the system scales.

Ultimately, such research is essential for designing adaptive autonomous systems capable of learning and cooperating effectively from structured guidance in high-stakes, real-world environments.

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